

GEN-3 SAFE HELICOPTER

DOCUMENT 00

10-TON / 7-ROTOR SAFETY-CENTERED BASELINE

Central Protected Safety Capsule Configuration

Seven equally sized 6.0 m rotors distributed around the central capsule • Four turboshaft-class prime movers

UPDATED CONCEPTUAL DESIGN DOCUMENT

Baseline update: 17 m × 17 m × 5 m overall target envelope

The main airframe body remains a nominal 4 m high envelope. The fixed safety capsule is positioned in the central heart of the vehicle and rises approximately 1 m above the main body, giving a nominal 5 m overall height target.

Rotor architecture is revised to seven equally sized 6.0 m rotors distributed around the protected central capsule. No primary lift rotor is located directly above the capsule.

Controlled item	Updated baseline	Status
Rotor configuration	7 × 6.0 m distributed rotors around central capsule	[T]
Central roof rotor	None	[T]
Overall envelope	17 m × 17 m × 5 m	[T]
Main body height	4 m	[T]
Capsule rise above body	≈ 1 m	[T]
External landing gear	6 fixed mechanical legs with thick spring/damper elements	[T]
Safety capsule	Fixed / non-jettisonable / centrally protected	[T]

PROPRIETARY CONCEPT INFORMATION

Concept / Pre-Feasibility Baseline — Not Certified. This document is not a type certificate, flight manual, structural substantiation package, production drawing set, or proof of achievable performance.

1. DOCUMENT CONTROL

Item	Controlled value
Document number	00
Revision	00-T10-R7-CPC-Full
Project	GEN-3 Safe Helicopter
Product baseline	10 metric tonnes / 7-rotor compact platform
Architecture	Central protected fixed capsule + seven equally sized distributed rotors
Rotor configuration	7 × 6.0 m nominal rotors
Total geometric disk area	≈197.9 m ²
Capsule configuration	Fixed, non-jettisonable
Fuel planning capacity	1,700 L nominal target
Prime movers	4 × turboshaft-class
Installed power target	2.5–2.8 MW
Overall envelope	17 m × 17 m × 5 m target
Main body height	4 m target
Central capsule rise	≈1 m above main body target
External landing gear	6 fixed mechanical spring/damper legs
Occupant capacity	7 (1 pilot + 6 passengers)
Status	Concept / pre-feasibility; engineering validation pending

STATUS DISCIPLINE

All numerical values remain targets, planning values, calculated estimates, assumptions, or open items unless supported by accepted evidence. This revision changes the controlled architecture and envelope; it does not convert conceptual values into verified results.

Baseline authority: this revision supersedes the prior 17 m × 17 m × 4 m geometry and the prior central-roof-rotor arrangement for the fixed-capsule baseline.

2. EXECUTIVE DESIGN STATEMENT

GEN-3 is a 10-ton-class distributed-lift rotorcraft concept organized around a centrally located protected occupant capsule. The revised architecture deliberately removes the primary lift rotor from directly above the capsule and distributes all seven equal-diameter rotors around the capsule perimeter.

The design intent is to create physical separation between the highest-consequence occupied volume and the primary rotor system while retaining distributed lift, control authority, modular propulsion channels, layered energy absorption, fire containment, water survival provisions, and aircraft-level emergency recovery.

The main body remains approximately 4 m high. The central capsule rises approximately 1 m above the main body, creating the 5 m overall envelope. The elevated capsule is treated as a structural and aerodynamic subsystem requiring dedicated analysis rather than as a cosmetic cockpit fairing.

The revised baseline is intentionally conservative in its claims. Rotor interference, induced-flow interaction, structural mass, center-of-gravity effects, control authority, drag and crash loads remain open engineering questions.

Primary design objectives

- Protect a defined occupant survival volume at the geometric and structural center of the vehicle.
- Avoid placing a primary lift rotor directly above the capsule.
- Use seven equally sized 6.0 m rotors for commonality and distributed lift.
- Use six fixed external mechanical landing legs with substantial spring/damper elements for energy absorption.
- Maintain a four-prime-mover / seven-rotor-channel distributed propulsion direction.
- Preserve the ten-layer safety architecture without granting certification credit before evidence exists.
- Maintain staged development from subscale tests to unmanned demonstrator and progressive flight testing.

3. FROZEN / CONTROLLED PRODUCT BASELINE

Parameter	Baseline	Status
Maximum design mass	10,000 kg	[T]
Rotor count	7	[T]
Rotor diameter	6.0 m each	[T]
Rotor arrangement	All seven around central capsule; no central roof rotor	[T]
Total rotor disk area	≈197.9 m ²	[C]
Disk loading at 10 t	≈50.5 kg/m ²	[C]
Fuel capacity	1,700 L nominal	[T]
Fuel reference mass	≈1,360 kg at 0.80 kg/L	[C-A]
Prime movers	4 × turboshaft-class	[T]
Power class	≈650–700 kW class/unit	[T]
Installed power target	2.5–2.8 MW	[T]
Length	17 m	[T]
Width	17 m	[T]
Main body height	4 m	[T]
Overall height	5 m	[T]
Capsule rise	≈1 m above main body	[T]
Capsule	Fixed / non-jettisonable	[T]
External landing legs	6 fixed mechanical spring/damper legs	[T]
Occupants	7	[T]
Aircraft recovery	8-parachute concept	[T]
Safety architecture	Ten-layer concept	[T]

LOCK RULE

These values are controlled design targets and calculated estimates. They shall not be silently modified. Any proposed change requires formal configuration control.

4. GEOMETRY AND CONFIGURATION DEFINITION

The updated vehicle envelope is 17 m long × 17 m wide × 5 m overall high. The 4 m main body envelope carries the primary distributed rotor structures, propulsion interfaces, landing-leg attachment structures, fuel zones, systems and capsule support interfaces.

The safety capsule occupies the central heart of the aircraft. Its upper geometry extends approximately 1 m above the main body envelope. The capsule is not intended to be a forward cockpit pod; it is a centrally located protected occupant volume.

The seven 6.0 m rotors form a surrounding distributed-lift arrangement. Exact radial spacing, azimuthal positions, longitudinal offsets and vertical separation remain subject to aerodynamic, structural and control trade studies.

Geometry item	Controlled concept	Open engineering question
Main body	17 × 17 × 4 m target	Detailed outer mold line and internal volume
Capsule	Central; ±1 m above body	Shell geometry, volume, glazing, access
Overall height	5 m target	Dynamic clearance and aerodynamic integration
Rotors	7 × 6.0 m	Exact spacing and interference minimization
Rotor plane	Distributed around capsule	Common plane vs staggered elevation
Landing gear	6 fixed legs below body	Attachment points, stroke and loads
Ground clearance	[AV]	Must follow rotor/structure/egress requirements

5. CENTRAL PROTECTED SAFETY CAPSULE

The capsule is the defining occupant-protection subsystem. It is fixed and non-jettisonable in this baseline. It is centrally positioned rather than placed at the nose, and its elevated upper section provides a dedicated protected volume while maintaining separation from the distributed rotor ring.

Conceptual features retained from the prior baseline

- Near-spherical / highly rounded external form for multi-directional impact load management.
- Thick multi-angle carbon-fiber roll-cage connecting structural regions around the occupied volume.
- Seven soft damper-mounted occupant seats, subject to final layout.
- Multiple high-stroke mechanical isolation mounts between capsule and main structure.
- Protected egress routes designed for loss of normal electrical power.
- Fire containment and suppression around the capsule perimeter.
- Controlled flotation and water-survival provisions.
- Defined survivability envelope instead of unsupported terms such as “crash-proof.”

The one-meter capsule rise is a design target, not evidence that the capsule is safer. It creates potential benefits and new risks: increased exposed surface area, altered center of gravity, possible drag increase, rotor wake interaction, structural overturning moments and different emergency-egress geometry.

6. SEVEN-ROTOR DISTRIBUTED-LIFT ARCHITECTURE

All seven primary rotors are nominally identical at 6.0 m diameter. No primary rotor is located directly above the central capsule. The rotors are distributed around the capsule so that the occupied volume remains physically separated from the primary lift disks.

Item	Concept value
Rotor count	7
Rotor diameter	6.0 m
Radius	3.0 m
Area per rotor	$\approx 28.27 \text{ m}^2$
Total geometric disk area	$\approx 197.9 \text{ m}^2$
10 t disk loading	$\approx 50.5 \text{ kg/m}^2$
15 t illustrative lift case	$\approx 75.8 \text{ kg/m}^2$
Central roof rotor	None
Rotor commonality	Same nominal diameter and modular station philosophy

Aerodynamic questions requiring closure

- Rotor-to-rotor induced-flow interference.
- Rotor-to-body and rotor-to-capsule interference.
- Tip clearance and ground-effect behavior.
- Yaw, pitch and roll authority after single-channel loss.
- Unequal inflow caused by the raised central capsule.
- Noise, vibration and acoustic interaction.
- Forward-flight drag and lift distribution.

7. ROTOR AERODYNAMIC SIZING AND POWER SANITY CHECKS

The following are first-order conceptual checks only. They do not constitute a validated rotor model.

Metric	Indicative value	Status
Weight at 10,000 kg	≈98.1 kN	[C]
Thrust per rotor at equal sharing	≈14.0 kN	[C]
Illustrative 20% total thrust margin	≈117.7 kN total	[C-A]
Per-rotor thrust at 20% margin	≈16.8 kN	[C-A]
Ideal induced hover power floor	≈1.40 MW	[C-A]
Approx. real hover power using FM=0.72	≈1.94 MW	[C-A]
Installed power target	2.5–2.8 MW	[T]

Important limitation

The seven-rotor surrounding arrangement may alter the idealized induced-power estimate through interference. The raised central capsule can also alter local inflow. The 1.94 MW estimate must be re-run after rotor coordinates, body outer mold line and capsule geometry are defined.

Required next analyses: actuator-disk screening, blade-element/momentum model, rotor wake interaction model, CFD, sensitivity to rotor spacing, and subscale and full-scale testing.

8. PROPULSION AND PRIME-MOVER ARCHITECTURE

The baseline retains four turboshaft-class prime movers feeding a segmented high-voltage electrical distribution architecture with seven independent rotor drive channels.

Item	Baseline direction	Status
Prime movers	4	[T]
Power class	≈650–700 kW/unit	[T]
Installed power target	2.5–2.8 MW	[T]
Power flow	Turboshaft → generator → segmented HV bus → inverter → rotor motor	[T]
Rotor channels	7 independent channels	[T]
Isolation	Individual rotor and prime-mover isolation	[T]
Final OEM selection	Open	[AV]
Generator matching	Open	[AV]
Certification / export rights	Open	[AV]

Redundancy shall be physical where safety credit is claimed. Shared buses, common cooling loops, common software, common sensors or common structural members can create common-cause failures and shall not be assumed independent without analysis.

The distributed architecture shall be tested under single prime-mover loss, single rotor-channel loss, power-bus faults, sensor faults, thermal degradation and combinations identified by the system safety process.

9. FUEL, ENERGY CONTAINMENT AND THERMAL SAFETY

Fuel capacity remains 1,700 L nominal. The fuel system is treated as a high-consequence energy source and must be separated from the protected occupant volume.

- Segregated fuel containment away from the capsule.
- Emergency fuel shutoff and zone isolation.
- Fire-resistant routing and protected lines.
- Leak detection and controlled venting.
- Thermal shielding between hot machinery and fuel.
- Drainage and fire-safe architecture after hard landing.
- Electrical isolation for damaged power distribution.

Reference fuel mass of approximately 1,360 kg at 0.80 kg/L is a planning calculation only. Final density, usable fuel, unusable fuel, thermal expansion, tank geometry and structural installation mass remain supplier- and design-dependent.

Thermal management must be designed around actual engine/generator/motor/inverter efficiencies and heat rejection requirements. The 2.5–2.8 MW installed-power target is not itself a thermal design closure.

10. MASS BUDGET AND WEIGHT CONTROL

The 10,000 kg figure remains a target, not a mass closure. The architecture update may increase capsule structural and support mass while reducing some rotor-central support complexity. A new bottom-up mass statement is required.

Mass group	Prior planning value	Status after update
Primary airframe / rotor support	2,050 kg	Re-baseline required
Safety capsule / occupant protection	1,150 kg	Re-baseline required
Seven rotors / hubs / control	720 kg	Re-baseline required
Powerplant / generators / drives	1,550 kg	Planning only
Fuel system excluding fuel	230 kg	Planning only
Fuel at 1,700 L	≈1,360 kg	Calculated planning value
Avionics / flight control / distribution	520 kg	Planning only
Landing gear	430 kg	Re-baseline for six legs
Fire / emergency / safety	540 kg	Planning only
Thermal management	350 kg	Planning only
Interconnects / hardware / reserve	1,200 kg	Planning only

No passenger or payload allowance is claimed from any residual mass until the updated bottom-up mass statement closes.

11. SIX-LEG MECHANICAL LANDING GEAR

The updated baseline uses six fixed mechanical landing legs located beneath the main body. Each leg concept includes thick spring/damper elements intended to absorb ground-impact energy before it reaches the primary structure and protected capsule.

Feature	Baseline concept
Quantity	6
Attachment	Main body primary structure
Leg type	Fixed mechanical
Energy absorption	Thick spring/damper elements
Tip / ground interface	High-load replaceable interface concept
Redundancy objective	No single leg failure shall directly imply capsule loss; requires analysis
Load cases	Level, one-leg overload, asymmetric, hard landing, side load
Verification	Component characterization + full-scale dynamic tests

“Thick spring” is a design descriptor, not a finalized stiffness specification. Spring rate, damping coefficient, stroke, load capacity, buckling resistance, thermal behavior and fatigue life remain [AV].

A six-leg arrangement changes ground-load distribution and may affect tip-over stability. Landing-gear geometry must be co-designed with center of gravity, rotor clearance, capsule egress and emergency landing loads.

12. MULTI-STAGE OCCUPANT ENERGY MANAGEMENT

The safety philosophy retains a sequential energy-management chain. The objective is to prevent high-energy loads from reaching the occupant survival volume as a single abrupt impulse.

Stage	Function
1. Landing legs	Initial ground-impact energy absorption
2. Outer / primary structure	Controlled load redistribution and deformation
3. Capsule mounts	Mechanical isolation and stroke
4. Capsule shell	Load spreading around occupied volume
5. Roll-cage	Occupant-volume retention and local load distribution
6. Seats / restraints	Final occupant acceleration attenuation

Each stage requires measurable load, displacement, rate and failure criteria. Multiple layers do not automatically provide additive safety credit. Common-mode structural failure can defeat multiple layers simultaneously.

Required future evidence includes component characterization, sled/drop testing, structural dynamic analysis, finite-element correlation, full-scale capsule tests and occupant-injury assessment using applicable certification and research criteria.

13. TEN-LAYER SAFETY ARCHITECTURE

Layer	Function	Baseline implementation
1	Occupant survival capsule	Central fixed capsule with multi-angle roll-cage
2	Multi-stage energy absorption	Six landing legs → structure → capsule mounts → shell → seats
3	Capsule isolation	Multiple high-stroke mechanical damper mounts
4	Distributed propulsion	Seven independent rotor channels
5	Prime-mover redundancy	Four turboshaft-class units
6	Fire detection / isolation / suppression	Zone detection, fuel/electrical isolation, suppression
7	Control redundancy	Segregated sensing, computing and actuation paths
8	Emergency descent / recovery	Eight-parachute aircraft-level concept + degraded modes
9	Water survival	Flotation, controlled leakage, protected egress
10	Structural containment / hazard isolation	Hazard containment and modular rotor isolation/jettison concept

No layer is a guaranteed survival mechanism. The safety case must quantify failure probabilities, loads, deployment envelopes, occupant injury criteria, environmental limits and test evidence before certification credit is claimed.

14. FIRE, THERMAL AND HIGH-ENERGY HAZARD CONTROL

- Thermal sensing in engine, generator, fuel, rotor-drive and high-energy electrical zones.
- Fire detection logic separated from propulsion control where practical.
- Fuel isolation valves and emergency shutoff.
- Fire-resistant barriers around occupied volume.
- Suppression around capsule perimeter and critical machinery zones.
- Power reduction and controlled load shedding.
- Post-fire safe-state logic.
- Containment and drainage designed to limit propagation.

The revised central capsule creates a different fire-zone geometry from the prior central-roof-rotor architecture. Fire analysis must be rebuilt around the seven surrounding rotor stations, four prime movers, fuel zones and central capsule.

Potential common-cause paths include shared electrical distribution, cooling, control networks, structural penetrations, fuel routing and thermal radiation. These shall be explicitly represented in FHA/FMEA/FMECA/FTA/SSA work.

15. FLIGHT CONTROL AND FAULT RECONFIGURATION

The seven distributed rotors provide multiple control effectors, but the exact controllability benefit remains an analysis problem. The control system must identify rotor thrust, speed, vibration, temperature, power and actuator anomalies and reconfigure within validated limits.

- Independent monitoring for each rotor channel.
- Commanded-versus-measured performance cross-checking.
- Automatic isolation of failed channels where safe.
- Redistribution of collective and attitude authority.
- Pilot warning hierarchy for degraded modes.
- Independent emergency power for flight-critical control functions.
- Hardware-in-the-loop and fault-injection testing.

The raised central capsule may change aerodynamic moments and local inflow. Control-law development must use final geometry, rotor coordinates and mass properties rather than a generic seven-rotor model.

16. STRUCTURAL LOAD PATHS AND CAPSULE INTERFACES

The main airframe forms a perimeter around the central capsule. Rotor, engine, landing, emergency-recovery and capsule loads are treated as distinct but interacting load paths.

- Dedicated local load paths at each rotor station.
- Primary structural ring around the central capsule.
- Controlled capsule-to-airframe interfaces with defined stroke and force limits.
- Six landing-leg attachment nodes tied into primary structure.
- Dedicated inspection and test requirements for high-consequence joints.
- Limit and ultimate load cases with load redistribution.

The 1 m elevated capsule introduces a potential overturning moment and vertical load path that must be explicitly modeled. It must not be assumed that the existing 4 m body structure can simply carry the additional geometry without reinforcement.

Certification principles may inform development, but no certification compliance is claimed by this document.

17. EMERGENCY RECOVERY CONCEPT

Scenario	Primary response	Secondary response	Validation
Single rotor loss	Control reconfiguration	Power redistribution / landing mode	Simulation + flight test
Single prime-mover loss	Power reallocation	Degraded-flight mode	HIL + flight test
Multiple rotor degradation	Emergency flight mode	8-parachute concept	Failure analysis + deployment test
Major fire	Detection / isolation / power reduction	Emergency landing / recovery	Fire + system tests
Severe structural event	Maintain capsule integrity	Aircraft recovery concept	Dynamic structural test
Water impact	Capsule survivability + egress	Flotation / recovery	Water-impact test

The eight-parachute concept remains a conceptual aircraft-level recovery provision. Exact canopy sizing, deployment envelope, reefing strategy, attachment loads, redundancy and certification credit remain [AV].

18. WATER SURVIVAL AND DITCHING

The central capsule is intended to provide a protected occupant survival volume during water-impact scenarios. The concept includes structural protection, controlled flotation, protected egress and independent emergency power.

- Watertight or controlled-leakage boundaries around the occupied volume.
- Flotation provisions sized after mass-property and stability analysis.
- Emergency egress accessible after defined vehicle attitudes.
- Independent emergency electrical power.
- Survival-equipment location and activation logic.
- Capsule interfaces designed to limit water-impact shock transmission.

The 5 m overall height and raised capsule change water-entry geometry and center-of-buoyancy considerations. Ditching analysis must therefore be updated rather than inherited from the previous 4 m configuration.

Water survivability is a major validation item and cannot be established from conceptual geometry alone.

19. MANUFACTURING AND PROTOTYPING STRATEGY

The revised architecture is intended to support staged physical development. The central capsule, rotor modules and six-leg landing system should be developed as testable subsystems before full aircraft integration.

- Phase 1: material coupons, composite roll-cage samples, spring/damper characterization.
- Phase 2: subscale rotor, motor, inverter, controller and thermal tests.
- Phase 3: full-scale capsule shell and dynamic impact articles.
- Phase 4: six-leg landing gear dynamic test rig.
- Phase 5: integrated multi-rotor propulsion test rig.
- Phase 6: unmanned integrated demonstrator.
- Phase 7: progressive flight-envelope expansion.
- Phase 8: certification-oriented conformity and compliance program.

The goal is not to prove the complete aircraft by documentation. Each high-consequence claim should acquire evidence through appropriately scaled physical test or accepted analysis.

20. VERIFICATION AND VALIDATION PLAN

Subsystem	Analysis	Bench	Dynamic	Flight
Rotor / hub	Yes	Yes	Yes	Yes
Propulsion channel	Yes	Yes	Yes	Yes
Flight controls	Yes	HIL	Fault injection	Yes
Capsule structure	FEA	Component	Full-scale impact	Integrated
Capsule mounts	Model	Characterization	Full-scale	Integrated
Six landing legs	Load model	Component	Full-scale dynamic	Integrated
Fire protection	Hazard analysis	Component	System fire test	As required
Emergency recovery	Simulation	Component	Deployment test	Flight demo
Water survival	Simulation	Tank tests	Water impact	As required

Verification traceability shall link each requirement to an objective method and evidence record. No target shall be upgraded to demonstrated performance without appropriate evidence.

21. CERTIFICATION AND REGULATORY DEVELOPMENT

The certification basis is not fixed by this concept document. Final aircraft category, maximum takeoff mass, operational concept, jurisdiction, occupant capacity and propulsion architecture will determine the applicable regulatory framework.

- Maintain a certification-assumption register.
- Identify applicable structural, rotor, propulsion, flight-control, occupant-protection and emergency-recovery requirements.
- Map each requirement to a planned means of compliance.
- Identify where analysis must be supported by component, subcomponent or full-scale test.
- Maintain authority engagement before treating any requirement as formally accepted.

No regulatory compliance or certification status is claimed in Document 00.

22. SYSTEM SAFETY ASSESSMENT FRAMEWORK

- Functional Hazard Assessment for flight-critical and occupant-protection functions.
- Preliminary System Safety Assessment for propulsion, flight control, fire, emergency recovery and capsule systems.
- FMEA/FMECA for rotor stations, prime movers, power conversion and landing gear.
- Fault Tree Analysis for loss of lift, loss of control, fire propagation and capsule compromise.
- Common-cause and common-mode failure analysis.
- Failure propagation analysis between fuel, hot machinery, electrical systems and capsule.
- Structural failure and load-path assessment.
- Human factors and emergency egress assessment.
- Fault detection, annunciation and crew workload assessment.

The ten layers are not ten independent guarantees. Shared dependencies may defeat multiple layers simultaneously. The safety case must therefore analyze common-mode vulnerabilities, independence assumptions and latent failures.

23. PRELIMINARY RISK REGISTER

Risk	Impact	Current treatment	Next closure action
Mass growth above 10 t	High	Growth reserve retained	Bottom-up mass + supplier data
Rotor interference increases power	High	6 m rotor concept	Rotor interaction CFD + test
Raised capsule affects stability	High	Central placement	CG / aero / control model
Six-leg load distribution inadequate	High	Spring/damper concept	Dynamic landing tests
Capsule structure too heavy	Medium/High	Dedicated allocation	FEA + material coupons
Common-mode control failure	High	Distributed channels	SSA + HIL fault injection
Fire propagation	High	Containment + isolation	Hazard analysis + tests
Parachute recovery load exceedance	High	8-canopy concept	Deployment simulation + structural test
Water-impact instability	High	Central capsule concept	Water-impact modeling + test
Thermal mass growth	Medium	Dedicated allocation	Thermal model + rig test
Engine/generator mismatch	High	Early OEM engagement	Matching study + supplier data

Risk levels are qualitative because no approved project risk matrix is defined in this document. Numerical risk scores remain [AV].

24. OPEN ENGINEERING ITEMS BEFORE NEXT REVISION

- Freeze detailed mission profile: payload, crew, cruise speed, altitude, range and endurance.
- Define exact seven-rotor coordinates and vertical offsets around the raised capsule.
- Complete rotor aerodynamic model including interference and forward-flight behavior.
- Define rotor RPM, blade geometry, solidity, tip speed and control method.
- Complete OEM engine and generator selection and matching.
- Close full power budget including hover, climb and degraded modes.
- Develop generator, motor, inverter and thermal models.
- Replace planning mass allocations with component-level estimates.
- Define capsule shell geometry, glazing, roll-cage load paths and egress.
- Quantify capsule mount stiffness, damping, stroke and force limits.
- Quantify six-leg spring/damper characteristics and hard-landing loads.
- Define parachute sizing, deployment envelope and attachment loads.
- Define water-impact and flotation requirements quantitatively.
- Create formal system-safety hazard and failure-mode database.
- Map final design to the applicable certification basis.

The next controlled revision should be triggered by closure of the rotor aerodynamic geometry, initial structural sizing loop, capsule energy-absorption model, six-leg landing model, OEM power package and controlled component-level mass budget.

25. DESIGN RULES FOR THE UPDATED BASELINE

Rule	Controlled design principle
1	Do not allow the 10,000 kg target to become a paper-only number; every major subsystem needs a measurable mass allocation.
2	Do not claim safety credit for emergency devices before deployment, structural and occupant-survivability evidence exists.
3	Prefer passive mechanical safety layers where they reduce dependence on software or electrical power.
4	Avoid common-mode dependencies that can defeat multiple safety layers simultaneously.
5	Treat the central capsule as a complete safety system, not simply as a stronger cabin.
6	Keep all seven rotor diameters nominally common unless a controlled aerodynamic trade proves a different architecture superior.
7	Do not place a primary lift rotor directly above the protected capsule in this baseline.
8	Treat the 1 m capsule elevation as a structural/aerodynamic design parameter, not a safety claim.
9	Use six landing legs as an energy-management system requiring quantitative characterization.
10	Maintain explicit separation between targets, calculations, simulations, tests and verified results.

26. PRELIMINARY PERFORMANCE / FEASIBILITY FRAME

Metric	Current conceptual treatment	Status
Gross mass	10,000 kg target	[T]
Rotor diameter	6.0 m × 7	[T]
Total disk area	≈197.9 m ²	[C]
Hover thrust requirement	≈98.1 kN total	[C]
Equal-share thrust	≈14.0 kN/rotor	[C]
Illustrative 20% thrust margin	≈117.7 kN total	[C-A]
Ideal induced hover power floor	≈1.40 MW	[C-A]
Approx. real hover power	≈1.94 MW using FM=0.72	[C-A]
Installed power target	2.5–2.8 MW	[T]
Endurance	[AV] pending updated mission and efficiency model	[AV]
Range	[AV] pending aerodynamic / mission model	[AV]
Cruise speed	[AV]	[AV]
Single rotor loss capability	Design objective only	[T]
Single prime-mover loss capability	Design objective only	[T]

The previous order-of-magnitude endurance and range estimates are intentionally not carried forward as controlled values because the rotor arrangement and 5 m envelope have changed. They must be re-derived after updated geometry and power models are closed.

27. FINAL UPDATED BASELINE STATEMENT

Document 00 Revision 00-T10-R7-CPC-Full establishes the updated GEN-3 compact fixed-capsule baseline as a 10-ton-class, seven-rotor distributed-lift rotorcraft with a centrally located protected occupant capsule.

The controlled geometry is 17 m × 17 m × 5 m overall. The main body is a 4 m nominal envelope and the central safety capsule rises approximately 1 m above the main body. The capsule is fixed, non-jettisonable and located in the central heart of the vehicle.

The lift architecture consists of seven equally sized nominal 6.0 m rotors distributed around the central capsule. There is no primary lift rotor directly above the capsule. The seven-rotor architecture remains coupled to four turboshaft-class prime movers and a segmented electrical distribution concept supplying seven rotor channels.

The external landing system is revised to six fixed mechanical legs with thick spring/damper elements beneath the main body. These legs are part of the occupant energy-management architecture and require quantitative stiffness, damping, stroke, strength and dynamic-test closure.

All prior safety architecture remains active: central occupant survival capsule, multi-stage energy absorption, capsule isolation, distributed propulsion, prime-mover redundancy, fire detection/isolation/suppression, control redundancy, aircraft-level emergency recovery, water survival and structural/hazard containment.

The updated architecture is a controlled design baseline, not a demonstrated aircraft. Rotor interference, power, mass closure, structural loads, capsule survivability, landing dynamics, emergency recovery, water survival, certification basis and flight performance remain subject to analysis and test.

FINAL ENGINEERING PRINCIPLE

ENGINEERING TRUTH > DOCUMENT COMPLETENESS

EVIDENCE > ASSERTION

CONFIGURATION CONTROL > SILENT CORRECTION

TRACEABILITY > CONVENIENCE

Controlled record	Value
Document	00
Revision	00-T10-R7-CPC-Full
Configuration ID	GEN3-SH-10T-FC
Status	Concept / Pre-Feasibility
Baseline authority	This revision supersedes the prior fixed-capsule geometry for the updated program baseline

Founder / Principal Author: Amirbahador Atai

END OF DOCUMENT 00